

Two Catalogs. More in Common Than You'd Think.

A ***semantic*** comparison of **23,948 courses** across **Arizona State** and the **University of Tennessee, Knoxville**.

Prepared for ASU Online Leadership · v2, April 2026 · Method: sentence-transformer embeddings

We were reading the catalogs wrong.

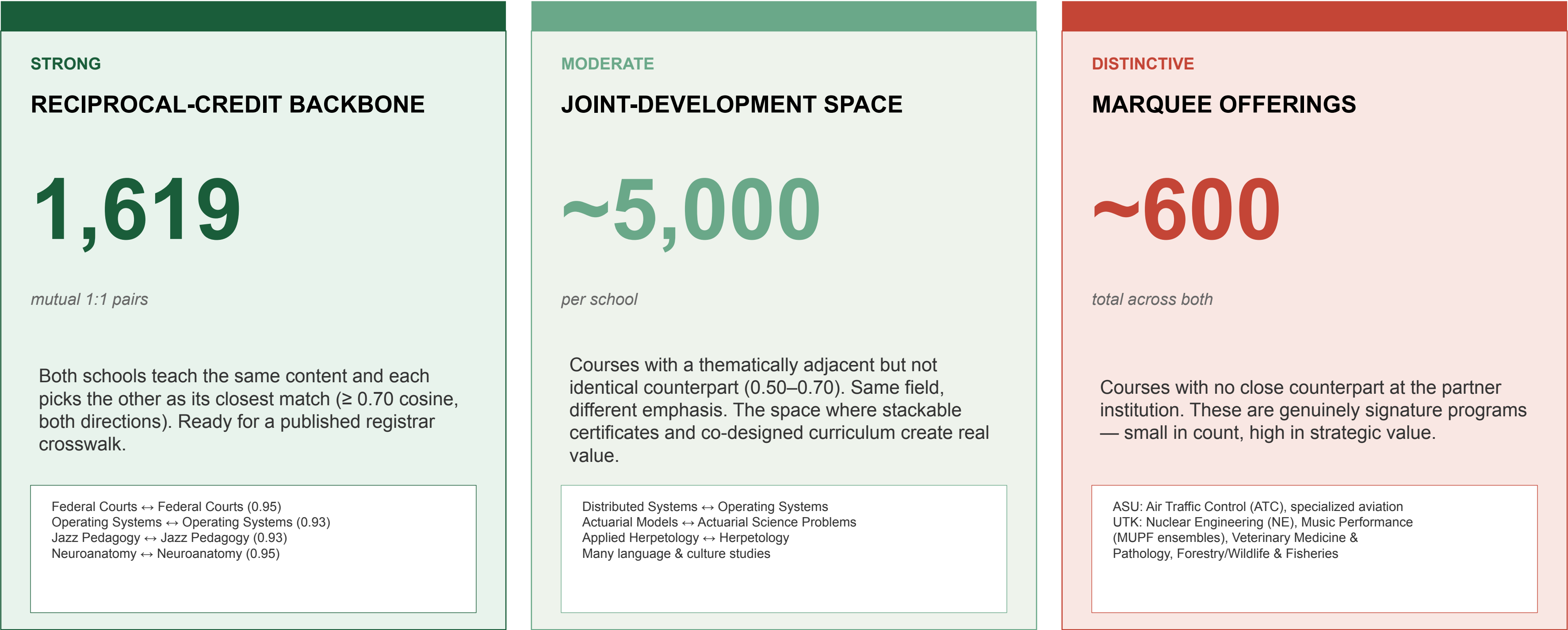


An initial lexical comparison (TF-IDF) suggested most of each catalog was unique to its school. Replacing it with a sentence-transformer model (semantic similarity on descriptions) shows the opposite: the catalogs overlap substantially, and the interesting story is in the middle layer — courses that teach similar things, differently.

v1 · Lexical (TF-IDF)	
Mutual-strong 1:1 pairs	316
Courses with strong match	~840
Catalog flagged as unique	62%
Narrative	“mostly distinct”

v2 · Semantic (sentence-transformer)	
Mutual-strong 1:1 pairs	1,619
Courses with strong match	~7,000
Catalog truly distinctive	~3%
Narrative	three layers: deep overlap + joint-dev + marquee

Three layers, three strategies.



Three different partnership strategies. The deck that follows walks each one.

1,619 clean 1:1 pairs.

Sample of highest-similarity mutual-strong pairs. Both sides describe the same content and each picks the other as its closest match. The low-friction foundation for automatic credit recognition.

SIM.	ASU	UTK
0.95	LAW 613 Federal Courts	LAW 916 Federal Courts
0.95	NEU 426 Neuroanatomy	NEUR 200 Introductory Neuroanatomy
0.93	CSE 330 Operating Systems	COSC 361 Operating Systems
0.93	MUE 560 Jazz Pedagogy	MUJZ 420 Jazz Pedagogy
0.92	MAE 587 Radiation Heat Transfer	ME 613 Advanced Radiation Heat Transfer
0.92	THP 517 Stage Management	THEA 330 Stage Management
0.92	ABS 260 Fundamentals of Sustainable Horticulture	PLSC 210 Horticulture: Principles and Practices
0.92	LAP 371 Landscape Architecture Professional Practice I	LAR 582 Professional Practices
0.91	MAT 443 Introduction to Abstract Algebra	MATH 351 Introduction to Abstract Algebra
0.89	SLC 515 Second-Language Acquisition	LING 476 Second Language Acquisition
0.86	ARB 102 Elementary Arabic II	ARAB 122 Elementary Arabic II
0.85	LAW 516 Criminal Law	LAW 809 Criminal Law

Full list of 1,619 pairs published alongside this deck.

Where reciprocal credit is already dense.

ASU subjects with the highest share of strong cross-school pairs (min. 10 non-shell courses). Each maps to a single dominant UTK subject. These are the clearest places to start a published crosswalk — agreement is already implicit in course content.

HEB	Hebrew	88% strong	→ UTK HEBR
LAP	Landscape Architecture	82% strong	→ UTK LAR
AES	Asian Studies	75% strong	→ UTK AFAS
CAP	Counseling Psychology	74% strong	→ UTK COUN
RTH	Recreation Therapy	73% strong	→ UTK RSM
POR	Portuguese	69% strong	→ UTK PORT
GRK	Greek (Classical)	68% strong	→ UTK CLAS
CED	Counselor Education	67% strong	→ UTK COUN
MIC	Microbiology	65% strong	→ UTK MICR
JST	Jewish Studies	65% strong	→ UTK REST

**Where each school
has distinctive depth.**

ASU's signature depth — not taught at UTK.

Subjects ranked by distinctiveness (unique + moderate share). Each has a meaningful course count and no strong counterpart at UTK.

ATC Air Traffic Control 10 courses · 70% distinctive · 10% moderate Tower Ops, TRACON, IFR Ground School	DCE Dance 132 courses · 10% distinctive · 80% moderate Partner: UTK THEA
BMD Biomedical Diagnostics 19 courses · 26% distinctive · 63% moderate Anatomic, Cytology, Pathology studies	HCA Health Care Admin. 10 courses · 10% distinctive · 90% moderate No direct UTK counterpart

Six more ASU distinctive subjects (ATC runner-ups in aviation, biomimicry, health administration) detailed in the HTML report drill-down.

UTK's signature depth — not taught at ASU.

The concentration of distinctiveness at UTK is sharper: Music Performance, Nuclear Engineering, and the land-grant agricultural sciences dominate.

MUPF Music Performance 243 courses · 49% distinctive · 47% moderate Ensembles, instrumental studios, conducting. Conservatory-grade.	NE Nuclear Engineering 110 courses · 31% distinctive · 58% moderate Radiochemistry, nuclear thermal hydraulics. ASU has no equivalent.
PLSC Plant Sciences 100 courses · 17% distinctive · 54% moderate Land-grant depth: cultivation, horticulture, plant breeding	ANSC Animal Sciences 56 courses · 23% distinctive · 55% moderate Large-animal husbandry, reproduction, nutrition
VMP Vet. Medicine Pathology 71 courses · 16% distinctive · 79% moderate	FORS Forestry 38 courses · 16% distinctive · 63% moderate

Different bets on what gen-ed is for.

ASU tags general-ed along traditional discipline axes. UTK's VolCore leads with pedagogy — Engaged Inquiries is tagged more widely than any discipline category. A real divergence that creates room for a co-designed credential.

ASU GENERAL STUDIES (GOLD)		
2,455 courses tagged		
G	Global awareness	575
SB	Social-Behavioral Sci.	524
HU	Humanities	507
L	Literacy & Critical Inquiry	440
H	Historical Awareness	322
C	Cultural Diversity	294

UTK VOLUNTEER CORE (VOLCORE)		
1,113 courses tagged		
EI	Engaged Inquiries	232
AH	Arts & Humanities	191
GCI	Global Citizenship — Intl.	154
AOC	Applied Oral Communication	96
NS	Natural Sciences	84
WC	Written Communication	80
UTK's top tag is pedagogical (Engaged Inquiries), not disciplinary.		

Six ways to work together.



FROM THREE LAYERS TO SIX OPPORTUNITIES

Six opportunities the data supports.

01 Reciprocal-credit compact

Publish a crosswalk for the 1,619 mutual-strong pairs. Removes transfer-evaluation friction.

EVIDENCE

1,619 pairs ≥ 0.70 cosine, both directions

FAST

02 Humanities + language stackable minor

Co-designed minor built from 10 ASU subjects with >65% strong overlap (HEB, LAP, AES, GRK, POR, JST).

EVIDENCE

HEB 88% · LAP 82% · AES 75% · GRK 68%

**JOINT
CREDENTIAL**

03 UTK-led applied-music certificate

Stackable UTK certificate delivered at ASU online scale. MUPF (243) + MUEN (63) as backbone.

EVIDENCE

MUPF 49% distinctive, 47% moderate

CONTENT IP

04 Land-grant × sustainability bridge

UTK FORS / PLSC / ANSC / VMP ↔ ASU ABS / SOS. Moderate content adjacency without duplication.

EVIDENCE

6 UTK subjects, ~350 combined courses

GRADUATE

05 Nuclear & aviation program swap

UTK NE (110 courses, 31% distinctive) and ASU ATC (10 courses, 70% distinctive) — articulation agreements.

EVIDENCE

Two highly-specialized, low-supply workforces

SPECIALTY

06 Applied-engaged micro-credential

Capstone tagged for UTK VolCore EI + ASU upper-div. L. Both institutional stamps on one credential.

EVIDENCE

UTK EI: 232 courses; ASU L: 440 courses

AMBITIOUS

Trust, but verify.

WHAT THE ANALYSIS IS

- Semantic embeddings**
sentence-transformers all-MiniLM-L6-v2, cosine similarity on title + description.
- Mutual-strong filter**
Both sides must pick each other at ≥ 0.70 cosine. The highest-confidence bucket.
- Shell bucket**
Thesis / Seminar / Internship / Special Topics filtered out of overlap counts heuristically.
- Catalog year note**
ASU + UTK-grad on 2025–26; UTK-undergrad on 2026–27. 3–5% churn typical.

WHAT TO DO NEXT

- Registrar review of 1,619 mutual pairs**
Fastest transfer of value from this analysis. Likely 90%+ accept-as-equivalent.
- Share with UTK Provost / Online Ed.**
Confirm appetite for 1–2 opportunities. Opp. 01 + Opp. 03 are the natural first pair.
- Model revenue for top-2 opportunities**
Build financial case. ASU online + UTK content IP arithmetic.
- Domain-tune the embedding model**
Fine-tune on catalog-description pairs to lift precision on the moderate↔strong boundary.

Let's build what neither school can build alone.



Built from a complete scrape of both live catalogs — **23,948 courses across 560 subject codes** — on 24 April 2026. Semantic matching with sentence-transformers.

Full drill-down available in the companion HTML report.